

Kodukäo nr 2

Õpilaskood: 244491; $a = 9+1=10$; $b = 1+1=2$

② a) $D(p) = 8ab - p^2$; $S(p) = 3bp^2 - a$

$$D(p) = 160 - p^2; S(p) = 6p^2 - 10$$

Tasakaalukind: $160 - p^2 = 6p^2 - 10$

$$7p^2 = 170 \quad | :7$$

$$p^2 = \frac{170}{7}$$

$$p^2 = \sqrt{\frac{170}{7}} \approx 4,928$$

b) 1) $D(p) = 160 - p^2 \Rightarrow D'(p) = -2p$

$$E_D = D'(p) \cdot \frac{p}{D(p)} = -\frac{2p^2}{160 - p^2}$$

$$E_D = -\frac{2 \cdot \frac{170}{7}}{160 - \frac{170}{7}} = -\frac{34}{95} \approx -0,358$$

$|E_D| < 1 \Rightarrow$ nõudlus on mitteelastne e jäik

2) $S(p) = 6p^2 - 10 \Rightarrow S'(p) = 12p$

$$E_S = S'(p) \cdot \frac{p}{S(p)} = \frac{12p^2}{6p^2 - 10}$$

$$E_S = \frac{12 \cdot \frac{170}{7}}{6 \cdot \frac{170}{7} - 10} = \frac{204}{95} \approx 2,147$$

$|E_S| > 1 \Rightarrow$ pakkumine on elastne

Soovid olla järgmine
superstaar DJ?

djs
djservice

Suurem valik
helis, valgus- ja
efektideid

$$c) \frac{\Delta Q}{Q} \approx E \cdot \frac{\Delta P}{P} \quad \frac{\Delta P}{P} = 2\% = 0,02$$

Nõudluse kogus: $\frac{\Delta Q_D}{Q} \approx (-0,358) \cdot 0,02 \approx -0,716\%$

~~Nõudluse~~
Pakutakse kogus: $\frac{\Delta Q_S}{Q} \approx 2,147 \cdot 0,02 \approx 4,294\%$

② $C(Q) = 0,16Q^2 + aQ$; $R(Q) = (7a - 0,26Q)Q$

$$C(Q) = 0,2Q^2 + 10Q$$
; $R(Q) = 70Q - 0,4Q^2$

$$\pi(Q) = R - C$$
; $\pi(Q) = 70Q - 0,4Q^2 - 0,2Q^2 - 10Q$

$$\pi(Q) = 60Q - 0,6Q^2$$

$$\pi'(Q) = 0$$
; $60 - 1,2Q = 0$
 $Q = 50$

$$\pi''(Q) = -1,2 < 0 \Rightarrow \text{maksimum}$$

a) $\frac{20a}{b} = \frac{20 \cdot 10}{2} = 100$

$$50 \leq 10 \Rightarrow \underline{Q^* = 50}$$

b) ~~$\frac{5a}{b}$~~ $\frac{5a}{b} = \frac{5 \cdot 10}{2} = 25$

$$50 > 25 \Rightarrow \underline{Q^* = 25}$$



③

$$y = \frac{6x^2}{20-x} ; y = \frac{2x^2}{20-x}$$

Nullstellen: $20 - x = 0$
 $x = 20$

1) $\lim_{x \rightarrow 20^+} \frac{2x^2}{20-x} = -\infty$

2) $\lim_{x \rightarrow 20^-} \frac{2x^2}{20-x} = \infty$

Kurvenansatz:

$$h = \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{20-x}}{x} = \lim_{x \rightarrow \infty} \frac{2x}{20-x} \stackrel{\frac{\infty}{\infty}}{\nearrow} \text{L'H} = \lim_{x \rightarrow \infty} \frac{2}{-1} = -2$$

$$b = \lim_{x \rightarrow \infty} \frac{2x^2}{20-x} + 2x \stackrel{(20-x)}{=} \lim_{x \rightarrow \infty} \frac{2x^2 + 40x - 2x^2}{20-x} \stackrel{\frac{\infty}{\infty}}{\nearrow}$$

$$= \lim_{x \rightarrow \infty} \frac{40}{-1} = -40$$

V: Nullstellen: $x = 20$; Kurvenansatz: $y = -2x - 40$